Medicinal Plant





Zanthoxylum chalybeum

15 natural products

LIGAND PREPARATION

Protonation at pH 5.0

RECEPTOR PREPARATION

Plasmepsin II (PMII)
Protonation at pH 5.0

MOLECULAR DOCKING

AutoDock Vina Simulation
15 natural products

DOCKING VALIDATION

R37 redocking

TOP DOCKING HITS

Docking score (kcal/mol)
THL: **–8.411**
Sesamin (SES): **–8.369**
Norchelerythrine (NCH): **–8.138**

MOLECULAR DYNAMICS

400 ns per system
Apo PMII | R37 | THL
SES | NCH

TRAJECTORY ANALYSIS

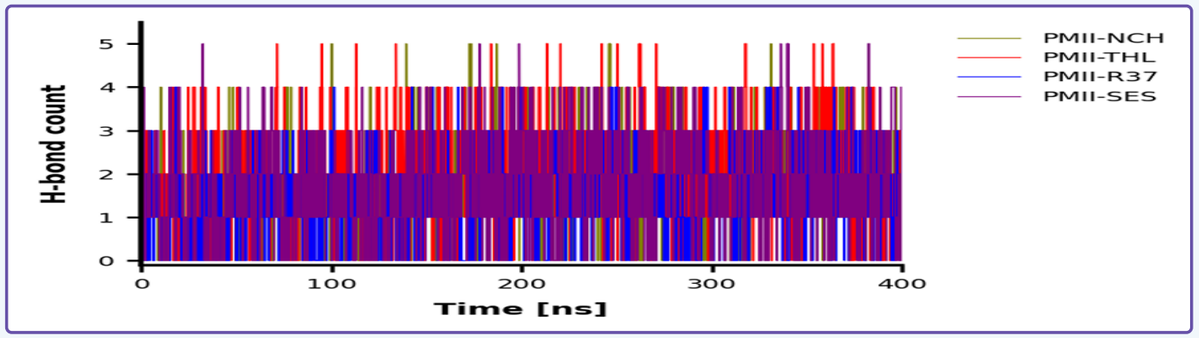
RMSD, RMSF, R_g, SASA
Binding-site retention
Hydrogen-bond dynamics

MM/PBSA ANALYSIS

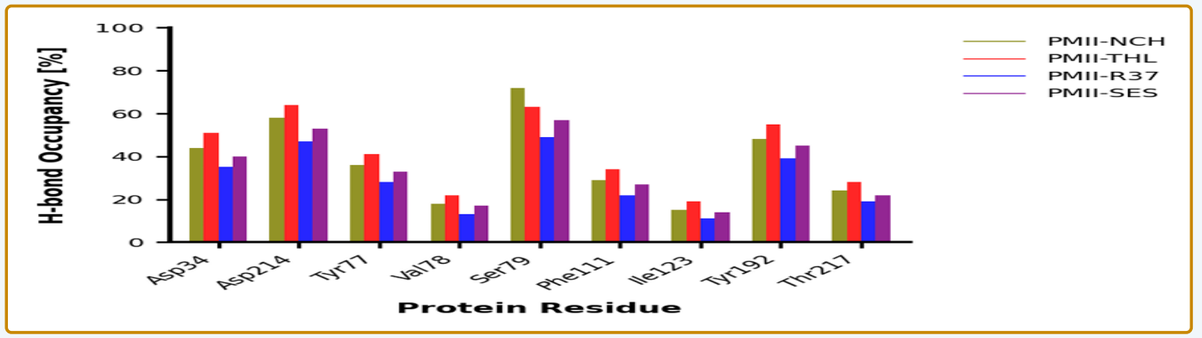
Final 200 ns (200–400 ns)
200 snapshots per complex
Binding-energy estimation

LEAD: 3,20,28-TRIHYDROXYLUPANE (THL)

H-BOND COUNT DYNAMICS (400 ns)



RESIDUE-SPECIFIC H-BOND OCCUPANCY



Docking score: **–8.411** kcal/mol • Contact retention: **82.8%** • Mean H-bonds: **1.93**
Asp34 occupancy: **51%** • Asp214 occupancy: **64%** • MM/PBSA: **–47.38** ± **2.72** kcal/mol